Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Computer Fundamentals & Applications

BCA — Semester 1 — 2019 — Answer Sheet

Subject Code: CACS101

Group A

Attempt ALL the questions. [10 × 1 = 10]

1. i) Which one of the following is not a non-impact printer?
a) Inkjet b) Dot-matrix c) Laser d) Thermal

ii) Which one of the following is the example of the utility software?
a) Anti-virus b) Operating system c) Language translators d) Word Processor

iii) Which one of the following is not a Network Operating System?
a) Unix b) Linux c) Windows 7 d) Windows Server

iv) Which one of the following key is used to uniquely identify a row in table?
a) Composite key b) Foreign Key c) Candidate Key d) Primary Key

v) Which one of the following layer of OSI reference model contains TCP protocol?
a) Network Layer b) Transport Layer c) Data Link Layer d) Application Layer

vi) Which one of the following is not a web server?
a) Apache b) IIS c) GWS d) HTTP

vii) Which one of the following is a crypto-currency?
a) Credit Card b) PayPal c) Bit Coin d) Mater Card

viii) Which one of the following command is used to shut down the computer normally?
a) shutdown /f b) shutdown /s c) shutdown /h d) shutdown /r

ix) What is the default font type in MS Word 2016?
a) Arial b) Times New Roman c) Calibri d) Microsoft Sans Serif

x) Which one of the following key is used for help in MS-PowerPoint?
a) F1 b) F5 c) F6 d) F11

[10 × 1 = 10]

Answer: Answers:
i) b) Dot-matrix — Dot-matrix is an impact printer (uses physical pins striking an ink ribbon). Inkjet, Laser, and Thermal are all non-impact printers.

ii) a) Anti-virus — Utility software helps maintain and optimize computer performance. Anti-virus, disk cleanup, and backup tools are utilities. OS is system software, language translators are development tools, and word processors are application software.

iii) c) Windows 7 — Windows 7 is a desktop/client operating system, not a Network Operating System (NOS). Unix, Linux, and Windows Server are NOS designed to manage network resources.

iv) d) Primary Key — A Primary Key uniquely identifies each row in a table. It cannot contain NULL values and must be unique.

v) b) Transport Layer — TCP (Transmission Control Protocol) operates at the Transport Layer (Layer 4) of the OSI model, providing reliable, connection-oriented communication.

vi) d) HTTP — HTTP is a protocol (HyperText Transfer Protocol), not a web server. Apache, IIS (Internet Information Services), and GWS are web server software.

vii) c) Bit Coin — Bitcoin is a decentralized digital cryptocurrency. Credit Card, PayPal, and MasterCard are traditional payment methods, not cryptocurrencies.

viii) b) shutdown /s — The command **shutdown /s** shuts down the computer normally. /f forces shutdown, /h hibernates, and /r restarts.

ix) c) Calibri — Calibri (11pt) is the default font in MS Word 2016 and other Office 2016 applications.

x) a) F1 — F1 is the universal Help key in Microsoft Office applications including PowerPoint. F5 starts slideshow, F6 navigates panes, F11 toggles full screen.

Group B

Attempt any SIX questions. [6 × 5 = 30]

2. What is Operating system? Explain the major functions of Operating System. [1+4]

[5]

Answer: Operating System (OS):
An Operating System is system software that acts as an intermediary between computer hardware and users. It manages hardware resources and provides services for computer programs. Examples: Windows, Linux, macOS, Android.

Major Functions of OS:

1. Process Management: Creates, schedules, and terminates processes. Allocates CPU time using scheduling algorithms (FCFS, SJF, Round Robin). Handles process synchronization and deadlock prevention.

2. Memory Management: Allocates and deallocates memory to programs. Tracks memory usage, implements virtual memory, and handles paging and segmentation.

3. File Management: Organizes files and directories. Manages file operations (create, read, write, delete). Controls file access permissions and maintains file system integrity.

4. Device Management: Manages I/O devices through device drivers. Allocates devices to processes and handles interrupts. Provides a uniform interface for diverse hardware.

5. Security and Protection: Authenticates users, controls access to resources, and protects against unauthorized access. Implements firewalls and encryption.

6. User Interface: Provides GUI (Graphical User Interface) or CLI (Command Line Interface) for user interaction.

3. Differentiate between primary and secondary memory. [5]

[5]

Answer: Difference between Primary and Secondary Memory:
Primary MemorySecondary Memory Directly accessible by CPUNot directly accessible by CPU Also called Main MemoryAlso called Auxiliary/External Memory Volatile (RAM loses data when power off)Non-volatile (retains data without power) Faster access speed (nanoseconds)Slower access speed (milliseconds) Smaller capacity (GB range)Larger capacity (TB range) More expensive per GBLess expensive per GB Examples: RAM, ROM, CacheExamples: HDD, SSD, DVD, USB Flash Stores currently running programs and dataStores permanent data, files, and backups
Primary Memory Types:
• **RAM (Random Access Memory):** Volatile, read/write, stores active programs
• **ROM (Read Only Memory):** Non-volatile, stores BIOS/firmware
• **Cache:** High-speed buffer between CPU and RAM

Secondary Memory Types:
• **Magnetic:** Hard Disk Drives (HDD)
• **Optical:** CD, DVD, Blu-ray
• **Flash:** SSD, USB drives, memory cards

4. What is computer virus? Explain symptoms of computer virus. [1+4]

[5]

Answer: Computer Virus:
A computer virus is a malicious software program designed to replicate itself and infect computer systems without the user's knowledge or consent. It attaches itself to legitimate programs or files and spreads when the infected program is executed.

Symptoms of Computer Virus:

1. Slow System Performance: The computer becomes noticeably slower than usual. Programs take longer to load and respond sluggishly.

2. Frequent Crashes: System crashes, freezes, or displays Blue Screen of Death (BSOD) unexpectedly.

3. Unusual Error Messages: Strange pop-ups, error messages, or dialog boxes appear without reason.

4. Missing or Corrupted Files: Files disappear, become corrupted, or their sizes change unexpectedly.

5. Increased Network Activity: Unexplained network traffic even when not actively using the internet, indicating the virus may be sending data.

6. Unauthorized Program Behavior: Programs open or close automatically, browser homepage changes, or new toolbars appear.

7. Disabled Security Software: Antivirus or firewall gets turned off without user action.

Prevention: Use updated antivirus, avoid suspicious downloads, keep OS patched, and backup regularly.

5. Define database? Explain the advantages of database over file based system. [1+4]

[5]

Answer: Database:
A database is an organized collection of structured data stored electronically in a computer system. It is managed by a Database Management System (DBMS) that allows users to create, retrieve, update, and manage data efficiently.

Advantages of Database over File-Based System:

1. Data Redundancy Control: In file systems, the same data is often duplicated across multiple files. Databases minimize redundancy through normalization, saving storage space and ensuring consistency.

2. Data Integrity: Databases enforce integrity constraints (primary keys, foreign keys, check constraints) to ensure data accuracy and validity. File systems have no such mechanism.

3. Data Security: DBMS provides robust security features including user authentication, access control, and encryption. File systems offer limited security.

4. Data Independence: Databases separate logical data structure from physical storage. Changes in storage don't affect applications. File systems tightly couple data and programs.

5. Concurrent Access: Multiple users can access and modify data simultaneously in databases without conflicts, using locking and transaction management.

6. Backup and Recovery: Databases have built-in mechanisms for backup, restore, and crash recovery. File systems require manual backup procedures.

7. Query Language: Databases support SQL for powerful data retrieval and manipulation. File systems require writing custom programs for each operation.

6. What is proxy server? Write down the benefit of using proxy server in the organization. [5]

[5]

Answer: Proxy Server:
A proxy server is an intermediary server that sits between client devices and the internet. It receives requests from clients, forwards them to the destination server, and returns the response to the clients. It acts as a gateway, masking the client's identity and controlling access.

Benefits of Using Proxy Server in Organization:

1. Enhanced Security: Proxy servers hide internal IP addresses, making it harder for attackers to directly target organizational systems. They can filter malicious content and block harmful websites.

2. Access Control: Administrators can restrict access to certain websites (social media, gambling, adult sites) during work hours, improving productivity and maintaining professional standards.

3. Bandwidth Savings: Proxy servers cache frequently accessed web content. When multiple users request the same page, it is served from cache instead of downloading repeatedly, saving bandwidth.

4. Anonymity and Privacy: Employees can browse the internet without revealing their actual IP addresses, protecting their privacy and preventing tracking.

5. Content Filtering: Proxies can scan incoming content for viruses, malware, and inappropriate material before it reaches the internal network.

6. Load Balancing: Reverse proxies distribute incoming traffic across multiple servers, improving performance and reliability.

7. Logging and Monitoring: Proxy servers maintain logs of internet usage, helping organizations monitor compliance and investigate security incidents.

7. Define e-commerce? Mention the benefits of using it in the context of customer? [1+4]

[5]

Answer: E-Commerce:
E-Commerce (Electronic Commerce) refers to buying and selling of goods and services over the internet. It includes online retail, electronic payments, online auctions, and internet banking. Examples: Amazon, Daraz, eSewa.

Benefits of E-Commerce for Customers:

1. Convenience: Customers can shop 24/7 from anywhere without visiting physical stores. No need to worry about store timings, parking, or travel.

2. Wider Product Selection: Online stores offer vast product catalogs from around the world. Customers can compare prices, features, and reviews across multiple sellers easily.

3. Better Prices: E-commerce reduces overhead costs (rent, staff) for sellers, allowing them to offer lower prices. Customers can easily find discounts, coupons, and deals.

4. Time Saving: Customers can find products quickly using search and filters. Checkout is fast with saved payment methods and addresses.

5. Detailed Product Information: Online listings include specifications, images, videos, and customer reviews, helping buyers make informed decisions.

6. Easy Returns and Refunds: Most e-commerce platforms offer hassle-free return policies with home pickup, reducing purchase risk.

7. Personalized Recommendations: AI algorithms suggest products based on browsing history and preferences, enhancing the shopping experience.

8. Consider the following structure and answer the questions. (E:/> is current prompt)
a) Write down DOS command to create above files and directories. [2]
b) Write down DOS command to move all text file into dir_3 directory. [1]
c) Write DOS command to delete dir_1 directory. [1]
d) Hide the root directory from E drive. [1]

[5]

Answer: Solution:
a) DOS commands to create files and directories:
E:\> md dir_1 E:\> md dir_2 E:\> md dir_3 E:\> cd dir_1 E:\dir_1> copy con file1.txt [Type content and press Ctrl+Z, Enter] E:\dir_1> copy con file2.txt [Type content and press Ctrl+Z, Enter]
Alternatively, using a single batch approach for creating a tree structure:
E:\> md dir_1 dir_2 dir_3 E:\> echo. > dir_1 ile1.txt E:\> echo. > dir_1 ile2.txt E:\> echo. > dir_2 ile3.txt
b) Move all text files into dir_3:
E:\> move *.txt dir_3E:\> move dir_1*.txt dir_3E:\> move dir_2*.txt dir_3\
c) Delete dir_1 directory:
E:\> del dir_1*.* /q E:\> rd dir_1
Or using the recursive delete command:
E:\> rd dir_1 /s /q
d) Hide the root directory from E drive:
E:\> attrib +h E:\
Note: Hiding the root directory is not recommended in practice as it may cause system issues. The attrib +h command sets the hidden attribute.

Group C

Attempt any TWO questions. [2 × 10 = 20]

9. a) From the following spreadsheet, write the formula to address the following conditions:
ABC Company, Kathmandu
Conditions:
1. Bonus will give 15% of salary if his/her salary is less than and equal to 6000. [1]
2. Tax will pay 10% of salary if his/her post is manager [1]
3. HA will get 5% of salary, if he/she is not from Ktm. [1]
4. Total is equal to the sum of (Salary, Bonus and HA) by deducting the Tax [1]
5. Write a formula to find the maximum salary above the excel sheet. [1]
b) What is Photoshop? Explain five major tools of Photoshop. [1+4]

[10]

Answer: a) Excel Formulas:

1. Bonus formula (Cell F2):
=IF(D2<=6000, D2*0.15, 0)
2. Tax formula (Cell G2):
=IF(C2="Manager", D2*0.10, 0)
3. HA formula (Cell H2):
=IF(B2<>"Ktm", D2*0.05, 0)
4. Total formula (Cell I2):
=D2+F2+H2-G2
5. Maximum Salary formula:
=MAX(D2:D4)
b) Photoshop:
Adobe Photoshop is a professional image editing software used for photo retouching, graphic design, digital art, and web design. It supports layers, masks, filters, and various tools for creative work.

Five Major Tools:
1. Move Tool: Moves layers, selections, and objects within the canvas. Essential for arranging elements in a design.

2. Marquee Tool: Creates rectangular, elliptical, or single-row/column selections. Used for selecting areas for editing, copying, or applying effects.

3. Brush Tool: Paints on the canvas with various brush styles, sizes, and opacities. Used for drawing, painting, and retouching with soft or hard edges.

4. Clone Stamp Tool: Copies pixels from one area and paints them over another. Ideal for removing blemishes, wrinkles, or unwanted objects from photos.

5. Magic Wand Tool: Selects areas of similar color with a single click. Useful for selecting backgrounds or solid-color regions for quick editing.

10. a) What is word processor? Explain the major functions available in the Home tab of word document. [5]
b) Define Network Topology? Write down the advantages of Network. [1+4]

[10]

Answer: a) Word Processor:
A word processor is a software application used to create, edit, format, and print text documents. Examples: Microsoft Word, Google Docs, LibreOffice Writer. It provides tools for typing, formatting, spell-checking, and document layout.

Major Functions in Home Tab of MS Word:

1. Clipboard Group: Cut, Copy, Paste, Format Painter — for moving and duplicating text and formatting.

2. Font Group: Change font family, size, color, bold, italic, underline, strikethrough, subscript, superscript, and text effects.

3. Paragraph Group: Alignment (left, center, right, justify), line spacing, bulleted/numbered lists, indentation, shading, and borders.

4. Styles Group: Apply predefined heading styles (Heading 1, Heading 2, Normal, Title) for consistent document formatting and automatic table of contents generation.

5. Editing Group: Find, Replace, and Select All — for quickly locating and modifying text throughout the document.

b) Network Topology:
Network topology refers to the physical or logical arrangement of devices (nodes) and connections (links) in a computer network. Common types: Bus, Star, Ring, Mesh, and Tree topology.

Advantages of Network:
1. Resource Sharing: Hardware (printers, scanners), software, and data can be shared among multiple users, reducing costs.

2. Communication: Email, instant messaging, video conferencing enable fast and efficient communication across distances.

3. Centralized Data Management: Data stored on central servers can be backed up, secured, and managed efficiently.

4. Cost Reduction: Shared resources and centralized administration reduce per-user hardware and maintenance costs.

5. Increased Storage Capacity: Network-attached storage (NAS) and cloud storage provide virtually unlimited storage accessible from any device.

11. a) What is animation? What are the advantages of using slide master in PowerPoint? [1+4]
b) Explain block diagram of computer and its various components. [5]

[10]

Answer: a) Animation:
Animation is the technique of creating the illusion of motion and change by rapidly displaying a sequence of static images (frames). In PowerPoint, animation refers to visual effects applied to text, images, and objects on slides to make presentations more engaging.

Advantages of Slide Master in PowerPoint:

1. Consistent Design: Slide master ensures all slides have uniform fonts, colors, backgrounds, and layouts throughout the presentation.

2. Time Saving: Changes made to the slide master automatically apply to all slides, eliminating the need to format each slide individually.

3. Easy Branding: Company logos, colors, and branding elements can be placed once in the master and appear on every slide.

4. Professional Appearance: Slide masters help create polished, professional-looking presentations with consistent alignment and spacing.

5. Footer and Page Numbers: Headers, footers, date, and slide numbers can be centrally managed and applied consistently.

b) Block Diagram of Computer:
A computer consists of five main components that work together:

1. Input Unit: Devices that feed data and instructions into the computer. Examples: Keyboard, Mouse, Scanner, Microphone.

2. Central Processing Unit (CPU): The brain of the computer consisting of:
• **ALU (Arithmetic Logic Unit):** Performs arithmetic and logical operations
• **CU (Control Unit):** Directs and coordinates all operations
• **Registers:** Small, fast storage locations for temporary data

3. Memory Unit: Stores data and instructions. Divided into Primary (RAM, ROM) and Secondary (HDD, SSD) memory.

4. Output Unit: Devices that present processed data to the user. Examples: Monitor, Printer, Speaker.

5. Storage Unit: Long-term data retention devices. Examples: Hard Disk, SSD, USB Drive, DVD.

Working: Data enters through Input → CPU processes using ALU/CU → Memory stores intermediate results → Output displays results → Storage saves permanently.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.